FitBuddy-AI Fitness Plan Generator using Gemini Models

Phase 1 – Brainstorming & Ideation

1. Introduction

The brainstorming and ideation phase is the starting point of the FitBuddy project. The main purpose of this phase is to identify a practical problem and convert the problem into a clear software idea. FitBuddy was conceived as an AI-powered web application that can generate personalized fitness plans based on basic information supplied by a user. Instead of providing the same generic workout routine to everyone, the proposed system considers details such as username, age, weight, fitness goal, and workout intensity.

The idea also includes nutrition guidance and a feedback mechanism. A user should not be limited to the first plan generated by the system. The user can provide feedback about the generated plan, after which the application can request an updated plan from the AI module. This makes the application more interactive and useful as a prototype of a personalized fitness assistant.

2. Problem Identification

Many users find it difficult to decide what type of workout routine is suitable for their personal goal. A beginner may not know how to organize exercises, while another user may want a different intensity or a different type of goal. Searching through multiple sources can also make the process confusing.

The project therefore identifies the need for a simple interface where users can enter a few basic details and receive an AI-generated plan. The system is intended as an educational software project demonstrating personalization, API integration, database storage, web development, and feedback-driven generation.

3. Proposed Solution

The proposed solution is FitBuddy, a web-based AI fitness plan generator. The frontend collects the required user information. FastAPI receives the submitted form data and validates it through the application schemas. The backend then communicates with the Gemini AI modules to generate the workout plan and nutrition tip. The generated information is stored in the database and displayed through a result page.

A second important part of the idea is plan improvement. If the user submits feedback, the application sends the original plan together with the feedback, goal, and age to the update module. The revised plan is then stored and displayed. Thus, the overall idea contains a generation stage and an improvement stage.

4. Brainstormed Features

The major features identified during ideation are personalized workout generation, nutrition tips, feedback-based plan revision, user data storage, existing-user update, administrator user management, user deletion, and a health-check endpoint. These features were selected because they demonstrate the main capabilities expected from an AI-enabled web application.

The project also requires a clear and simple user interface. The user should be able to enter information without dealing directly with the API implementation. At the same time, the project should expose FastAPI documentation through the Swagger interface so that the technical workflow can be demonstrated during project presentation.

5. Expected Benefits

FitBuddy is expected to demonstrate how generative AI can be connected with a conventional web application. The project combines user input, backend processing, AI generation, database storage, and feedback handling in one workflow. From an academic perspective, this gives the team practical

experience in Python, FastAPI, SQLAlchemy, HTML/CSS, API integration, and software testing.

The project is also designed to make the user experience straightforward: enter details, generate a plan, view the result, give feedback, and obtain an updated plan.

6. Initial Project Scope

The initial scope focuses on a working prototype rather than a complete medical or professional fitness platform. The application works with the user information required by the current project implementation and uses Gemini-based generation. It does not attempt to replace a qualified fitness or medical professional. The scope is therefore suitable for demonstrating software engineering and AI integration concepts.

7. Conclusion

The brainstorming and ideation phase established the central concept of FitBuddy and identified the problem, proposed solution, main features, expected benefits, and initial scope. The output of this phase provides the foundation for the next stage, where the functional and non-functional requirements of the system are analyzed in detail.

Phase Deliverables / Evidence

Item	Purpose	Status / Evidence
Problem statement	Defines the user problem	Documented
Core idea	Defines the AI fitness generator	Documented
Feature list	Defines initial scope	Documented